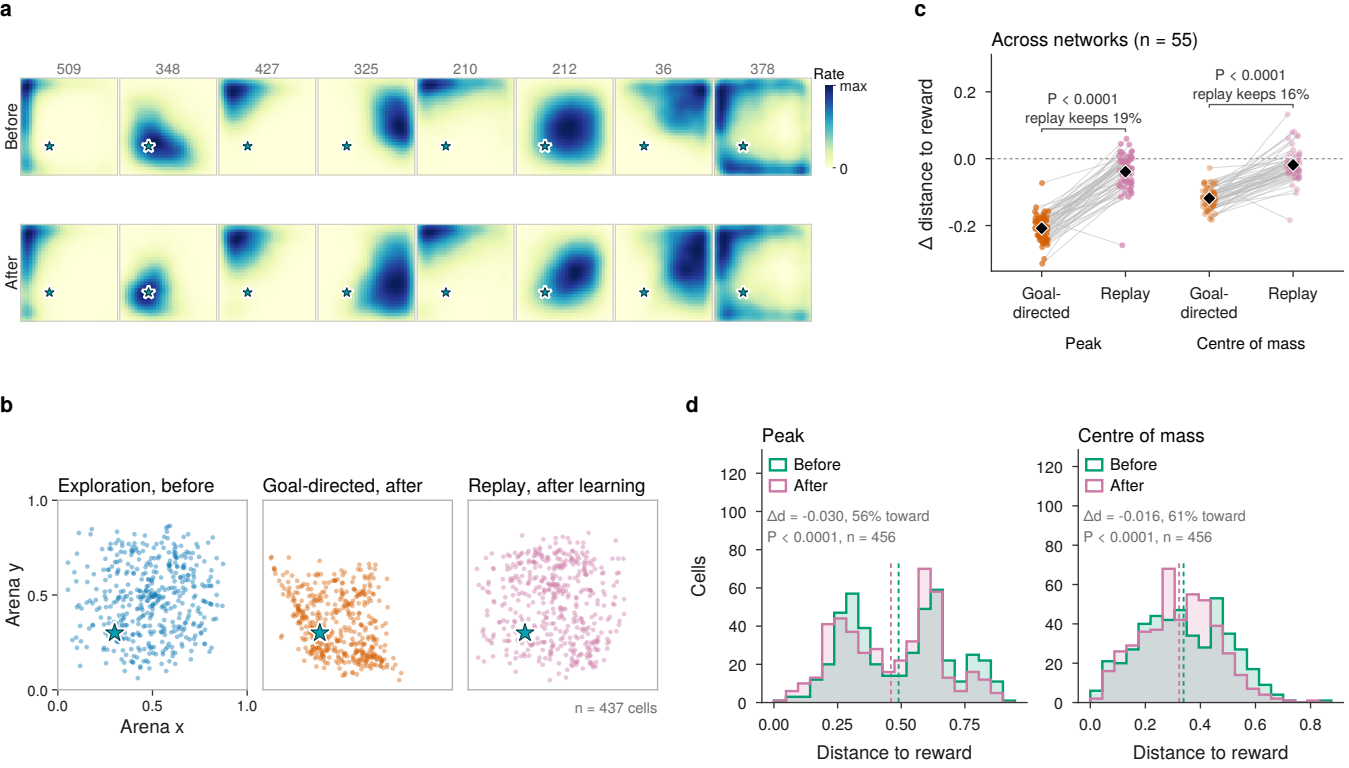

a
509
348
427
325
210
212
36
378
Rate
max
0
Before
After
c
Across networks (n = 55)
Δ distance to reward
0.2
0.0
-0.2
P < 0.0001
replay keeps 19%
P < 0.0001
replay keeps 16%
Goal-directed
Replay
Goal-directed
Replay
Peak
Centre of mass
b
Exploration, before
Goal-directed, after
Replay, after learning
Arena y
Arena x
1.0
0.5
0.0
0.0
0.5
1.0
n = 437 cells
d
Peak
Centre of mass
Before
After
Δd = -0.030, 56% toward
P < 0.0001, n = 456
Δd = -0.016, 61% toward
P < 0.0001, n = 456
Cells
120
100
80
60
40
20
0
0.00
0.25
0.50
0.75
0.0
0.2
0.4
0.6
0.8
Distance to reward
Distance to reward